Matteo Rizzuto

Postdoctoral Fellow · Department of Biology, Concordia University
Visiting Scientist · Department of Biology, Memorial University of Newfoundland

matteorizzuto.github.io | matteorizzuto | Matteo Rizzuto Pronouns: he, him, his

Highlights

- Ecosystem ecologist with 10+ years of experience in **process-based ecosystem modelling**, **animal ecology**, **spatial ecology**, and **biogeochemistry**.
- Published **15 peer-reviewed papers**, of which five as first author and two as co-first author, **with more than 300 combined citations and an h-index of 10**.
- Skilled in **translating scientific knowledge into practical insights** to inform stakeholders, and co-developing strategies to mitigate climate change and restore ecosystems.
- Experienced consultant in **ecosystem modelling applied to nature-based solutions** for the World Wildlife Fund, Conservation South Africa, Rewilding Europe, Fundación Rewilding Chile, and the Global Rewilding Alliance.
- **Invited speaker at the 2024 New York Climate Week** on how integrating animal controls on carbon cycling in nature-based solutions can help mitigate climate change and biodiversity loss.

Education

Doctor of Philosophy, Biology — *2016–2022*

Memorial University of Newfoundland — *St. John's, NL, Canada*

- Thesis title: From elements to landscapes: the role of terrestrial consumers in ecosystem functioning
- Advisor: Dr. Shawn J. Leroux
- Description: Designed and led a study of animal influences on the biogeochemistry of the Canadian Boreal Forest combining laboratory work, mathematical modelling, and intensive fieldwork in remote environments. Led a team of 5 graduate students and 5 research assistants.

Master of Research (Distinction), Ecology, Evolution, and Conservation — *2013–2014*

Imperial College London — *London, UK*

- Description: Developed and led two independent research projects investigating (i) the relationship of daily activity time with body size in carnivores and (ii) the performance of different methods to assess local species richness and biodiversity in the field.

Master of Science, Evolution of Animal and Human Behaviour — *2009–2012*

University of Turin — *Turin, TO, Italy*

- Description: Designed and led a study of predator-prey dynamics in the Western Italian Alps, combing strenuous, solitary fieldwork in remote alpine environments with extensive laboratory work and multivariate statistical analyses.

Bachelor of Science (Honours), Biology — *2004–2009*

University of Turin — *Turin, TO, Italy*

- Description: Studied the contributions of individual *Indri indri* lemurs to a family group's territorial song, to assess potential for sound-based, remote identification of focal individuals.

Research Positions

Postdoctoral Fellow — *2024–Present*

Department of Biology, Concordia University — *Montréal, QC, Canada*

- Leading a study on how species dispersal, competitive interactions, and landscape structure interact to shape the biodiversity-ecosystem function relationship. Remote appointment.
- Advisor: Prof. Pedro R. Peres-Neto

Visiting Scientist — *2024–Present*

Ecosystem Ecology Lab, Department of Biology, Memorial University of Newfoundland — *St. John's, NL, Canada*

- Conducting research on how species dispersal and landscape structure shape the biodiversity-ecosystem function relationship.

Postdoctoral Associate — *2022–2024*

Schmitz Lab, School of the Environment, Yale University — *New Haven, CT, USA*

- Developed a new modelling framework to quantify animal-mediated ecosystem carbon capture and inform nature-based solutions.

Visiting Assistant in Research — *Feb.–Apr. 2019*

Schmitz Lab, School of the Environment, Yale University — *New Haven, CT, USA*

- Developed a mathematical model exploring the effects of animal-mediated nutrient transport on ecosystem functions.

Research Assistant *Jan.–Mar. 2016*
Pawar Lab, Department of Life Sciences, Imperial College London *Silwood Park, Ascot, UK*
- Collected and managed data for a meta-analysis of plant thermal responses to climate change.

Research Assistant *Jun.–Oct. 2015*
Tsaobis Baboon Project, Zoological Society of London *London, UK*
- Collected behavioural, dietary, and population data from wild *Chacma* baboons in the Namib Desert.

Research Assistant *Jun.–Sept. 2013*
Roe & Red Deer in Trentino and Technology Project, Fondazione Edmund Mach *Trento, TN, Italy*
- Collect data on and created online maps of roe deer movement and plant phenology in the Eastern Italian Alps.

Graduate Intern *Jun.–Dec. 2011*
Piedmont Wolf Project, Maritime Alps Nature Park *Entracque, CN, Italy*
- Designed and led a lab- and field-based study of wolves-chamois predator-prey interactions in the Western Italian Alps.

Undergraduate Intern *2008–2009*
Ethology Lab, University of Turin *Turin, TO, Italy*
- Processed data for a study of family group songs among *Indri* lemurs.

Consultancy

Ecological consultant and lead modeller *2024–2025*
Conservation South Africa *St. John's, NL, Canada*
- Led a project assessing Common Eland effects on grassland carbon capture in uKhahlamba Drakensberg Park, South Africa, to inform management practices and human-wildlife relations. Authored an internal technical report summarizing key findings.

Ecological consultant and lead modeller *2024*
WWF Netherlands, Global Rewilding Alliance *New Haven, CT, USA*
- Assessed Forest Elephant effects on rainforest carbon capture to inform management regimes in Dja-Odzala-Minkébé and Sangha protected areas of the Congo Basin, Africa. Authored an internal technical report summarizing key findings.

Lead ecological modeller *2024*
Lincoln Institute of Land Policy *New Haven, CT, USA*
- Modelled the impacts of local predators and herbivores on ecosystem carbon capture in the tropical dry forest of the Yucatán Peninsula, Mexico. Authored an internal technical report summarizing key findings.

Lead ecological modeller *2024*
Fundación Rewilding Chile, Global Rewilding Alliance *New Haven, CT, USA*
- Quantified the potential additional carbon captured by a rewilded Puma-Guanaco system in Patagonia National Park, Chile. Authored an internal technical report summarizing key findings. In preparation for submission to a peer-reviewed scientific journal.

Ecological consultant and lead modeller *2024*
WWF Netherlands, WWF Romania, Rewilding Europe, Global Rewilding Alliance *New Haven, CT, USA*
- Quantified the potential additional carbon capture mediated by a rewilded European Bison population in the South Carpathian Mountains, Romania. Authored an internal technical report summarizing key findings and a scientific manuscript currently **in review** at *Ecological Monographs*.

Grants and Fellowships

Exploring wilder grasslands and the carbon offset opportunities using a "reanimated" carbon cycle model in South Africa. *Sept.–Dec. 2024*
Conservation South Africa *Value: US $ 20,000*
Role: principal investigator.

Publications

A † marks equal contributions. When listed as 2nd or 3rd author, I contributed to ideas generation, study design, data collection and analysis, interpretation, and writing. When listed as 4th author or later, I contributed to ideas generation, data collection, and writing.

Peer reviewed

15. Retez, G.†, Rizzuto, M.†, Kuemmerle, T., Schmitz, O. J. Exploring and mapping the potential effects of European bison rewilding on carbon dynamics in a mosaic landscape. Accepted, *Ecological Monographs*.

14. Rizzuto, M., Leroux, S. J., Schmitz, O. J. 2024. Rewiring the carbon cycle: a theoretical framework for animal-driven ecosystem carbon sequestration. *Journal of Geophysical Research: Biogeosciences* **129**, e2024JG008026. 10.1029/2024JG008026. ***Research Spotlight***
    Media coverage
    - Scharping, N. Animals deserve to be included in global carbon cycle models, too. *Eos* **105**. 16 April 2024.
    - Dinneen, J. Animals may help ecosystems store 3 times more carbon than we thought. *The New Scientist*. 19 April 2024.
13. Rizzuto, M., Leroux, S. J., Schmitz, O. J., Vander Wal, E., Wiersma, Y. F., Heckford, T. R. 2024. Animal-vectored nutrient flows across resource gradients influence the nature of local and meta-ecosystem functioning. *Ecological Modelling* **488**, 110570. doi: 10.1016/j.ecolmodel.2023.110570.
12. McLeod, A.M., Leroux, S.J., Rizzuto, M., Leibold, M.A., Schiesari, L. 2024. Integrating ecosystem and contaminant models to predict the effects of ecosystem fluxes on contaminant dynamics. *Ecosphere*, **15**(1): e4739. doi: 10.1002/ecs2.4739
11. Heckford, T.R., Leroux, S.J., Vander Wal, E., Rizzuto, M., Balluffi-Fry, J., Richmond, I.C., Wiersma, Y.F. 2022. Ecoregion and community structure influences on the foliar elemental niche of balsam fir (*Abies balsamea* (L.) Mill.) and white birch (*Betula papyrifera* Marshall). *Ecology and Evolution* **12**, e9244. doi: 10.1002/ece3.9244.
10. Little, C.J.[†], Rizzuto, M.[†], Luhring, T.M., Monk, J.D., Nowicki, R.J., Paseka, R.E., Stegen, J.C., Symons, C.C., Taub, F.B., Yan, J.D.L. 2022. Movement with Meaning: Integrating Information into Meta-Ecology. *Oikos* **8**, e08892. doi: 10.1111/oik.08892. ***Editor's Choice***
9. Balluffi-Fry, J., Leroux, S.J., Wiersma, Y.F., Richmond, I.C., Heckford, T.R., Rizzuto, M., Kennah, J.L., Vander Wal, E. 2022. Integrating plant stoichiometry and feeding experiments: state-dependent forage choice and its implications on body mass. *Oecologia* **198**(3), 579–591. doi: 10.1007/s00442-021-05069-5.
8. Richmond, I.C., Balluffi-Fry, J., Vander Wal, E., Leroux, S.J., Rizzuto, M., Heckford, T.R., Kennah, J.L., Riefesel, G.R., Wiersma, Y.F. 2022. Individual snowshoe hares manage risk differently: integrating stoichiometric distribution models and foraging ecology. *Journal of Mammalogy* **103**(1), 196–208. doi: 10.1093/jmammal/gyab130.
7. Heckford, T.R., Leroux, S.J., Vander Wal, E., Rizzuto, M., Balluffi-Fry, J., Richmond, I.C., Wiersma, Y.F. 2022. Spatially explicit correlates of plant functional traits inform landscape patterns of resource quality. *Landscape Ecology* **37**, 59–80. doi: 10.1007/s10980-021-01334-3.
6. Rizzuto, M., Leroux, S.J., Vander Wal, E., Richmond, I.C., Heckford, T.R., Balluffi-Fry, J., Wiersma, Y.F. 2021. Forage stoichiometry predicts the home range size of a small terrestrial herbivore. *Oecologia* **197**(2), 327–338. doi: 10.1007/s00442-021-04965-0. ***Winner, Hanski Prize 2021***
5. Ellis-Soto, D.[†], Ferraro, K.M.[†], Rizzuto, M., Briggs, E., Monk, J.D., and Schmitz, O.J. 2021. A methodological roadmap to quantify animal-vectored spatial ecosystem subsidies. *Journal of Animal Ecology* **90**(7), 1605–1622. doi: 10.1111/1365-2656.13538. ***Winner, Sidnie Manton Award 2021***
4. Richmond, I.C., Leroux, S.J., Vander Wal, E., Heckford, T.R., Rizzuto, M., Balluffi-Fry, J., Kennah, J., Wiersma, Y.F. 2021. Temporal variation and its drivers in the elemental traits of four boreal plant species. *Journal of Plant Ecology* **14**(3), 398–413. doi: 10.1093/jpe/rtaa103.
3. Balluffi-Fry, J., Leroux, S.J., Wiersma, Y.F., Heckford, T.R., Rizzuto, M., Richmond, I.C., Vander Wal, E. 2020. Quantity-quality trade-offs revealed using a multiscale test of herbivore resource selection on elemental landscapes. *Ecology and Evolution* **10**(24), 13847–13859. doi: 10.1002/ece3.6975.
2. Rizzuto, M., Leroux, S.J., Vander Wal, E., Wiersma, Y.F., Heckford, T.R., Balluffi-Fry, J. 2019. Patterns and potential drivers of intraspecific variability in the body C, N, and P composition of a terrestrial vertebrate, the snowshoe hare (*Lepus americanus*). *Ecology and Evolution* **9**(24), 14453–14464. doi: 10.1002/ece3.5880.
1. Rizzuto, M., Carbone, C., and Pawar, S. 2018. Foraging constraints reverse the scaling of activity time in carnivores. *Nature Ecology and Evolution* **2**(2), 247–253. doi: 10.1038/s41559-017-0386-1. ***Cover Story***
    Media coverage
    - John, J. Constantly on the hunt, midsize carnivores face unique risks. *The Wildlife Society*. 4 January 2018.

In progress

- Rizzuto, M., Espinoza, I., Saucedo, C., Schmitz, O. J. Trophic rewilding pumas and guanacos: estimating the potential to enhance carbon sequestration in a Patagonian grassland ecosystem. In review, *Ecological Solutions and Evidence*. Manuscript id: ESO-26-07-216.R1. bioRxiv preprint: 10.64898/2026.07.28.740956

- McLeod, A.M., Harris, H.A., Amyntas, A., Meyer, G.A., Peller, T., Rizzuto, M., Ferraro, K.M. The community-wide consequences of primary producer movement. In review, *Ecology Letters*. Manuscript id: 9029909.
- Ferentinos, A., Weeks, B., Berg, N., Boonstra, R., Cleveland, S., Emmel, A., Gese, E., Gigliotti, L., Hodges, K., Kielland, K., Kumar, A., Mills, L., Peelle, L., Rizzuto, M., Roerick, T., Sheriff, M., Siren, A., Underwood, H. B., Wirsing, A., Zimova, M. Snow and temperature shape snowshoe hare foot length under a changing climate. In review, *Journal of Animal Ecology*. Manuscript id: JAE-2026-00802.
- Ferraro, K., Meyer, G. A., Roman, J., Orr, D., Leroux, S., Subalusky, A., Duvall, E., Monk, J., Gewirtzman, J., Rizzuto, M., Koltz, A., Owens, C., Abraham, A., Doughty, C., Schmitz, O., Forbes, E. A Practical Primer for Zoogeochemistry: Essential Biogeochemical Methods for Animal Ecologists. In review, *Journal of Animal Ecology*. Manuscript id: JAE-2026-01006.
- le Roux, E., Subalusky, A., Kristensen, J., Abraham, A., Bakker, E., Bello, C., Berzaghi, F., Bump, J., Doughty, C., Dutton, C., Duvall, E., Ferraro, K., Forbes, E., Gelin, U., González, A. L., Hawlena, D., Jaafar, R., Joly, F.-X., Koltz, A., Larsen, A., Larsen, J., Leroux, S., Losada, M., McCary, M. A., Meyer, G. A., olde Venterink, H., Orr, D., Owens, C., Ribeiro, L., Rizzuto, M., Roman, J., Schmitz, O., Sobral, M., Ellis Soto, D., Svenning, J.-C., Twining, C., Trepel, J., Veldhuis, M., Wijas, B., Zanne, A., Villar, N., Schoelynck, J. Zoogeochemistry: the role of animals in biogeochemical cycles. In revision, *Functional Ecology*. Manuscript id: FE-2026-00145.R1.

Technical reports and Working papers

- Rizzuto, M. 2024. Animating the Carbon Cycle: Exploring wilder grasslands and carbon offset opportunities using a "reanimated" carbon cycle model in South Africa. *Conservation International South Africa*.
- Schmitz, O.J., Rizzuto, M., Retez, G. 2024. Animating the Carbon Cycle: Assessment of the feasibility of using European Bison to enhance ecosystem carbon capture and storage in the Carpathian Mountains of Romania. *WWF-Netherlands*.
- Schmitz, O.J., Rizzuto, M. 2024. Animating the Carbon Cycle: Assessment of the feasibility of using Guanaco and Puma to enhance ecosystem carbon capture and storage in Patagonia National Park, Chile. *Fundación Rewilding Chile*.
- Schmitz, O.J., Rizzuto, M. 2024. Animating the Carbon Cycle: Assessment of the potential role of Pumas, Jaguars, White-tailed deer, and Collard peccaries in controlling carbon capture and storage in Mexican Dry Tropical Forest Ecosystems. WP24OS1, *Lincoln Institute of Land Policy*.
- Schmitz, O.J., Rizzuto, M. 2024. Animating the Carbon Cycle: Assessment of the Potential Role of Forest Elephants in Controlling Carbon Capture and Storage in the TRIDOM Tropical Forest Ecosystem of the Congo Basin. *WWF-Netherlands*.

Outreach

- Wiersma, Y.F., Catto, N., Deal, C., Edinger, E., Evans, R., Geissinger, E., Hearn, C., Sun Lim, K., McCann, N., MacDonald, K., Meyer, A., Prosser, J., Quinn, D., Richmond, I.C., Rizzuto, M., Roncal, J., Swain, M. 2020. The classroom goes virtual—experiences at Memorial University. *Blog post*. url: https://nllandscapeecology.com.
- Cagnacci, F., Rocca, M., Nicoloso, S., Ossi, F., Peters, W., Mancinelli, S., Valent, M., Rizzuto, M., Hebblewhite, M. 2013. Il progetto 2C2T. *Il Cacciatore Trentino*, 93, 4–15. url: https://en.calameo.com/.

Presentations

Conference Talks

- Rizzuto, M., Retez, G, Schmitz, O.J. (2026, June 29–30) *Rewiring the cycle: an applied modeling approach to estimate potential carbon benefits of integrating animal influences on ecosystem biogeochemistry into nature-based solutions*. "Nature-based Solutions: Who benefits, who decides, who pays?" Symposium, organized by the British Ecological Society. York, UK.
- Rizzuto, M., Khattar, G., Peres-Neto, P.R. (2026, May 11–14) *From dispersal to productivity: assembling biomass in metacommunities.* 20th Annual Meeting of the Canadian Society for Ecology and Evolution, Toronto, Ontario, Canada.
- Rizzuto, M., Leroux, S.J., Schmitz, O.J. (2025, Jun. 29–Jul. 4) *Modeling the zoogeochemical effects of herbivores on ecosystem carbon cycles: theory and practice.* 61st Annual Meeting of the Association for Tropical Biology and Conservation, Oaxaca de Juàrez, Oaxaca, Mexico. ***Invited talk***

- Rizzuto, M., Leroux, S.J., Schmitz, O.J. (2023, Aug. 6–11) *Modeling the zoogeochemical effects of herbivores on ecosystem carbon cycles.* "For All Ecologists" Ecological Society of America Annual Meeting, Portland, OR, USA. ***Invited talk***
- Rizzuto, M., Leroux, S.J., Schmitz, O.J. (2023, May 25) *Modeling the zoogeochemical effects of herbivores on ecosystem carbon cycles.* 6th Yale Postdoc Association Annual Symposium, New Haven, CT, USA.
- Rizzuto, M., Leroux, S.J., Schmitz, O.J., Vander Wal, E., Wiersma, Y.F., Heckford, T.R. (2023, May 2–6) *Movers and shakers: Animal-vectored nutrient flows across resource gradients influence local and meta-ecosystem functioning.* "Ecological Models for Tomorrow's Solutions" The International Society for Ecological Modelling Global Conference 2023, Toronto, ON, Canada.
- Rizzuto, M., Leroux, S.J., Schmitz, O.J., Vander Wal, E., Wiersma, Y.F., Heckford, T.R. (2021, Aug. 2–6) *Going against the flow: non-diffusive organismal movement influences local and meta-ecosystem functioning.* "Vital Connections in Ecology" Ecological Society of America Virtual Annual Meeting, Long Beach, CA, USA.
- Rizzuto, M., Leroux, S.J., Vander Wal, E., Wiersma, Y.F., Heckford, T.R., Balluffi-Fry, J. (2018, Jul. 18–21) *Ontogeny and Ecological Stoichiometry of Snowshoe hares (Lepus americanus) in the Boreal Forests of Newfoundland.* Canadian Society for Ecology and Evolution Annual General Meeting, Guelph, ON, Canada.
- Rizzuto, M., Carbone, C., and Pawar, S. (2016, Jul. 7–11) *Bio-mechanical constraints on foraging reverse the scaling of activity rate among carnivores.* Canadian Society for Ecology and Evolution Annual General Meeting, St. John's, NL, Canada.

Conference Posters

- Rizzuto, M., Leroux, S.J., Vander Wal, E., Wiersma, Y.F., Heckford, T.R., Balluffi-Fry, J. (2018, Jul. 21–27) *Beyond Diffusion: Animal-Mediated Nutrient Transport at Different Spatial Scales.* "Unifying Ecology Across Scales" Gordon Research Seminar and Conference, Biddeford, ME, USA.

Workshops

- Rizzuto, M. (4 Nov. 2022) *Grammar of Graphics: ggplot2.* Part of SPRY: A Learning Community for Quantitative Skill-Sharing. School of the Environment, Yale University.
- Rizzuto, M. (14 Oct. 2022) *(R)markdown: a brief tour.* Part of SPRY: A Learning Community for Quantitative Skill-Sharing. School of the Environment, Yale University.

Outreach

- Rizzuto, M. (30 Apr. 2025) *How animals shape our world: ecosystems, nutrient cycling, and carbon capture.* Skype a Scientist, delivered to the Conservation & Conversations Discussion Group at the Louisville Zoo, Louisville, KY, USA.
- Rizzuto, M. (4 Dec. 2024) *What is an Ecosystem Ecologist anyway?* Skype a Scientist, delivered to an 8th grade class at Tri-West Middle School, Lizton, IN, USA.
- Rizzuto, M. (24 Sept. 2024) *Understanding the evidence. The latest science on trophic rewilding.* delivered at the New York Climate Week as part of the "Are wild animals the unsung heroes of climate action?" panel organized by the International Fund for Animal Welfare. ***Invited talk***
- Rizzuto, M. (26 Apr. 2024) *How animals shape our world: ecosystems, nutrient cycling, and carbon capture.* Wolfram Ecological Research Colloquium. ***Invited talk***
- Rizzuto, M. (16 Apr. 2024) *How animals shape our world: ecosystems, nutrient cycling, and carbon capture.* Skype a Scientist, delivered online as part of the Edward Jones Scientist Expo Week.
- Rizzuto, M. (22 Mar. 2022) *What is an Ecosystem Ecologist anyway?* Skype a Scientist, delivered to an 8th grade class at Hope Township Elementary School, Hope, NJ, USA.

Teaching

Memorial University of Newfoundland *2025*

Guest Lecturer *St. John's, NL, Canada*

Conservation Biology I: Conservation in Biology and Geography, Undergraduate course *Fall 2025*

- Designed and delivered a lecture on the diverse history, underlying philosophy, and approaches of Rewilding and Restoration

Southern Oregon University *2025*

Guest Lecturer (remote) *Ashland, OR, USA*

Biogeochemistry, Undergraduate course *Fall 2025*

- Delivered a lecture on animal effects on ecosystem nutrient cycling and carbon capture, and their potential as nature-based climate change solutions

Connecticut College *2024*

Guest Lecturer *New London, CT, USA*

Plant Ecology, Undergraduate course *Spring 2024*

- Delivered a lecture on plant-animal interactions and their effects on ecosystem biogeochemistry, functioning, and services

Yale University *2022–2024*

Guest Lecturer *New Haven, CT, USA*

The Modern Instructor, Lecture series *Fall 2023*

- Delivered two lectures on ecosystem ecology and climate change aimed at a generalist audience

Ecosystems and Landscapes, Graduate course *Fall 2023*

- Lead instructor on one of five course modules, focused on linking biodiversity to biogeochemistry at multiple spatio-temporal scales

Industrial Ecology, Graduate course *Fall 2022*

- Delivered a lecture bridging the metabolic theory of ecology, ecosystem ecology, and industrial ecology

Memorial University of Newfoundland *2017–2020*

Teaching Assistant and Guest Lecturer *St. John's, NL, Canada*

Models in Biology, Graduate and Undergraduate course *Winter 2020*

- Delivered three lectures introducing students to classical models in ecology, models of species interactions, and meta-ecological models

Principles of Ecology, Undergraduate course *Fall 2018*

- Managed the online learning part of the course, providing administrative and academic support to students

Graduate Core Seminar, Graduate course *Fall 2018*

- Delivered a seminar on Student-Supervisor Communications, focusing on how to establish effect and productive communication and how to resolve conflict

Principles of Biology, Undergraduate course *Winter 2017*

- Assisted students during lab-based lectures, marked lab reports, midterms exams, and invigilated final exams

Imperial College London *2014–2015*

Teaching Assistant *London, UK*

Ecology, Undergraduate course *Spring 2015*

- Demonstrator for the course's field trip, helped students design, collect data, and run analyses for their final projects

Behavioral Ecology, Undergraduate course *Winter 2015*

- Assisted with lab-based lectures, from experiment setup to helping students with lab data analysis

Introduction to Biological Statistics, Undergraduate course *Fall 2014*

- Helped students learn base and advanced R language, facilitated Q&A sessions ahead of final exams

Statistics, Graduate course *Fall 2014*

- Helped students in developing and coding statistical analyses in R

Macroecology and Climate Change, Graduate course *Fall 2014*

- Assisted students using the species distribution software Maxent

Student Supervision & Mentorship

Graduate supervision

- **Fall 2025-Winter 2026:** mentored a graduate student at Memorial University of Newfoundland, helping them to develop research ideas and methods for their MSc thesis
- **Spring 2024:** mentored a graduate student at Yale University in data analysis and visualization for their MSc thesis

Undergraduate supervision

- **Spring-Summer 2018:** supervised two undergraduate students at Memorial University of Newfoundland during

their USRA field-based research assistantship
- **Fall 2017:** supervised an undergraduate student during their lab-based research assistantship at Memorial University of Newfoundland , as part of the Memorial's Undergraduate Career Experience Program

Service

Editorial

Subject Matter Editor, General Ecology — *2024–Present*
Ecosphere — *Ecological Society of America*

Reviewer for *Nature Communications*, *Nature Ecology and Evolution*, *Soil Biology and Biochemistry*, *Journal of Environmental Management*, *Science of the Total Environment*, *Ecological Monographs*, *Journal of Advances in Modeling Earth Systems*, *Biological Conservation*, *Journal of Animal Ecology*, *Scientific Reports*, *Ecological Modelling*, *Royal Society Open Sciences*, *Ecology and Evolution*, *Trends Open*.

Expert reviewer — *2026*
UK Research and Innovation

Conference organization

6TH and 7TH Annual Postdoc Symposium — *2023–2024*
Yale University — *New Haven, CT, USA*

Community

Yale Postdoctoral Association — *2022–2024*
Yale University — *New Haven, CT, USA*
- Buddy Program Mentor, 2023–2024
- Symposium Committee Co-coordinator, 2023
- Symposium Committee Member, 2022

Biology Graduate Student Association — *2017–2020*
Memorial University of Newfoundland — *St. John's, NL, Canada*
- Communications officer, 2019–2020
- Chairperson, 2018–2019
- Seminar Co-coordinator, 2017–2018

Educator — *2000–2013*
Associazione O.A.S.I.—Operazione Mato Grosso di Torino — *Turin, TO, Italy*

Team Leader — *2005–2006*
XX Winter Olympic and IX Paralympic Games — *Turin, TO, Italy*

P Professional Development

Note: *Memorial University* stands for Memorial University of Newfoundland.

Courses

McCall MacBain Postdoctoral Leadership Certification — *13–17 July 2026*
McCall MacBain Postdoctoral Fellows Teaching and Leadership Program — *McMaster University*

Postdoctoral Mentorship Training — *March 2023*
Office for Postdoctoral Affairs — *Yale University*

Career Development Leaders Program — *Jan.–Mar. 2023*
Office of Career Strategy — *Yale University*

Becoming a More Equitable Educator: Mindsets and Practices — *Mar.–Jun. 2020*
MIT Teaching Systems Lab — *Attended online at edX.org*

Teaching Skills Enhancement Program — *Sept. 2019–Aug. 2020*
Centre for Innovation in Teaching and Learning — *Memorial University*

Workshops

IDP and Self-Exploration to Achieve Career Success — *14 & 26 February 2025*
School of Graduate Studies, Graduate Career Development — *Concordia University*

Advanced Teaching Workshops: writing the teaching statement — *14 September 2023*
Yale Poorvu Center for Teaching and Learning — *Yale University*

The Art of Teaching: balancing passive and active learning — *13 September 2023*
Yale Postdoc Association and Yale Poorvu Center for Teaching and Learning — *Yale University*

Inclusive Leadership Training — *30 Aug. 2023*
Yale Postdoc Association and Yale DEI Office — *Yale University*

Bystander Intervention Training — *13 Jun. 2023*
Yale Postdoc Association — *Yale University*

How to Help: tips from mental health first aid — *1 Mar. 2023*
Being Well at Yale — *Yale University*

Alt-Academic Career Q&A — *8-9 Feb. 2023*
Early Career Section — *Ecological Society of America*

How to Write an Effective Diversity Statement — *11 Oct. 2022*
Graduate Writing Lab — *Yale University*

Building Your Brand Workshop — *8 Sept. 2022*
Beyond the PhD — *Attended online*

Reproducible Research through Open Science — *11 Jun. 2020*
Canadian Institute for Ecology and Evolution & NSERC-CREATE "Living Data Project" — *Attended online at osf.io*

Teaching Inclusively & Equitably Online — *21 May 2020*
American Society for Engineering Education & NSF INCLUDES Aspire Alliance — *Attended online*

H5P Maker Session — *24 Oct. 2019*
Centre for Innovation in Teaching and Learning — *Memorial University*

Community of Inquiry Coffee Break: Open Access — *23 Oct. 2019*
Centre for Innovation in Teaching and Learning — *Memorial University*

Open Access and Scholarly Publishing — *22 Oct. 2019*
Centre for Innovation in Teaching and Learning — *Memorial University*

Four Things to Consider for Graduate Student Teaching — *8 Nov. 2018*
Enhanced Development of the Graduate Experience — *Memorial University*

AARMS CRG Software Carpentry Workshop — *27 May 2017*
The Carpentries — *Memorial University*

Honors and Awards

Postdoc Excellence Award, Quebec Centre for Biodiversity Science, Montréal, QC, Canada — *2026*
Hanski Prize, Oecologia, Springer Nature — *2021*
Mitacs Research Training Award, Mitacs, St. John's, NL, Canada — *2020*
Mitacs Globalink Research Award, Mitacs, St. John's, NL, Canada — *2019*
Dean's Doctoral Award, Memorial University of Newfoundland, St. John's, NL, Canada — *2016–2020*
Graduated with Distinction, Imperial College London, London, UK — *2014*
Erasmus-LLP Scholarship, University of Turin, Turin, Italy — *2010*

Professional Affiliations

British Ecological Society — *2026–2027*
Canadian Society for Ecology and Evolution — *2016–2026*
Association for Tropical Biology and Conservation — *2025–2026*
International Society for Ecological Modelling — *2023–2025*
Ecological Society of America — *2021–2023*